Introduction to Linux Service Theory TASK-1

1. **Install the Apache web server on your Linux machine using the package manager (apt or yum), and confirm the installation by running the command to check its version.**

Ans :

- Install Apache:
    - sudo yum install httpd -y
- Start and enable the service:
    - sudo systemctl enable --now httpd
- Verify the installation and check the version:
    - httpd -v
- Example output:
    - Server version: Apache/2.4.62
    - Server built: 2025-xx-xx

The installation is successful if the version command (apache2 -v or httpd -v) returns the Apache server version without errors.

2. **Start the Apache service using systemctl, then use systemctl status to verify it is running.**

Ans :

- Start Apache:
    - sudo systemctl start httpd
- Check the service status:
    - sudo systemctl status httpd
- Expected output includes:
    - httpd.service - The Apache HTTP Server
    - Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled)
    - Active: active (running)

- Quick Verification
    - sudo systemctl is-active httpd

**3. Stop and restart the Apache service using systemctl, and observe the difference between stop, start, and restart commands.

Hint: Use 'systemctl stop apache2', 'systemctl start apache2', and 'systemctl restart apache2' (or 'httpd' on some systems).**

Ans :

- Replace apache2 with httpd:
    - sudo systemctl stop httpd
    - sudo systemctl start httpd
    - sudo systemctl restart httpd
- Difference Between the Commands

Command	**What It Does**
systemctl stop apache2	Stops the Apache service completely. The web server becomes unavailable until started again.
systemctl start apache2	Starts the Apache service if it is not running.
systemctl restart apache2	Stops Apache and immediately starts it again. Commonly used after changing configuration files.

4. Open your browser and access http://localhost or your Linux machine's IP address to verify that the Apache default web page is displayed.

Ans :

- Open a web browser
  - http://localhost
- Verification from the Terminal
  - curl http://localhost

You should receive HTML content for the default Apache page.

5. Write a short comparison (3-4 lines) between Apache and Nginx web servers, mentioning one key use-case for each based on your research.

Ans :

Apache vs Nginx:

Apache	**Nginx**
Apache uses a process/thread-based architecture and is known for its flexibility and extensive module support.	Nginx uses an event-driven architecture, making it highly efficient at handling many concurrent connections.
Apache is often preferred for traditional web hosting environments, especially when applications require .htaccess support.	Nginx is commonly used as a reverse proxy, load balancer, and for serving high-traffic websites.
Key use-case: Hosting dynamic websites running PHP applications such as WordPress.	Key use-case: Serving static content and handling large volumes of web traffic efficiently.